

APP FEATURES & CREDENTIALS REFERENCE

Vishal Gold App — Complete Feature Inventory & Access Guide

Project	Vishal Gold — B2B/B2C Jewelry App
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1. Document Purpose

This document is the single authoritative reference for every feature implemented in the Vishal Gold application, covering both the Customer-facing app and the Administrator Panel. It records all system credentials, default login values, user roles, access levels, every screen and its capabilities, all backend services, and infrastructure details.

CONFIDENTIALITY: Sections containing credentials must be treated as strictly confidential. Store this document in a password-protected location and rotate any credentials shown here after the initial handover.

2. User Roles & Default Credentials

2.1 Role Matrix

Role	Auth Method	Access Level	Created By
CUSTOMER	OTP (Firebase Phone Auth)	Customer app only	Self-registration on first OTP login
WHOLESALER	OTP (Firebase Phone Auth)	Customer app + wholesale pricing	Self-registration; fills company details on first login
ADMIN	Email + Password (Firebase)	Full Admin Panel	Created by SUPER_ADMIN
SUPER_ADMIN	Email + Password (Firebase)	Admin Panel + sub-admin management	Provisioned manually in Firestore

2.2 Default Super Admin Credentials

IMPORTANT: These are the initial Super Admin credentials. Change the password immediately after first login.

Field	Default Value
Email	admin@vishalgold.com
Password	Vishal!@#456
Role	SUPER_ADMIN
Firestore Path	admins/{adminDocumentId}

2.3 Admin Account — Firestore Fields

Field	Type	Description
uid	String	Firebase Auth UID

email	String	Login email address
name	String	Display name on dashboard
role	String	'ADMIN' or 'SUPER_ADMIN'
createdAt	Timestamp	Account creation time
permissions	Map	Optional granular permission flags

2.4 Customer / Wholesaler Account — Firestore Fields

Field	Type	Description
uid	String	Firebase Auth UID
phone	String	Verified mobile number (E.164 format)
name / fullName	String	User display name
email	String	Optional email
role	String	'USER' or 'WHOLESALE'
company_name	String	Wholesaler business name
address	String	Wholesaler address
city	String	Wholesaler city
profile_image_url	String	Firebase Storage URL for profile picture
createdAt	Timestamp	First login timestamp
fcmToken	String	Latest FCM device token for push notifications
whatsapp	String	Optional WhatsApp number

2.5 Quick-Login PIN (Admin Only)

Setting	Detail
PIN length	6 digits
Storage	Device-local flutter_secure_storage (encrypted keystore on Android)
Fallback	Full email + password if PIN is forgotten or after 5 failed attempts
Biometric support	Available if device hardware supports fingerprint / face unlock
Scope	Device-specific — not synced to cloud

3. Authentication System

3.1 Customer OTP Flow (phone_auth_screen.dart)

1. Customer opens the app and is shown the Phone Auth screen.

2. Customer enters their 10-digit Indian mobile number.
3. Firebase sends a 6-digit OTP via SMS. Auto-read on Android (no reCAPTCHA).
4. Customer verifies the OTP to complete login.
5. On first login, a user document is auto-created in Firestore 'users' collection.
6. If WHOLESALER role is detected, the Company Details screen is shown next.

3.2 Admin Email + Password Flow (admin_login_screen.dart)

Security Feature	Implementation
Email + Password	Firebase Authentication EmailPassword provider
Role verification	Cross-checked against Firestore 'admins' collection on every login
Quick Login PIN	Optional 6-digit device-local PIN (quick_login_setup_screen.dart)
PIN Unlock Screen	pin_unlock_screen.dart — shown instead of full login if PIN is set
Session persistence	Firebase Auth token auto-refreshed across app restarts
Logout	Clears Firebase token + local PIN state

3.3 Auth-Related Screens

Screen File	Purpose
auth_screen.dart	Router that determines which auth path to show (admin vs customer)
phone_auth_screen.dart	Customer OTP login — phone number entry + OTP verification
admin_login_screen.dart	Admin email + password login form
pin_unlock_screen.dart	Quick unlock screen for admins who have enabled PIN login
setup_pin_screen.dart	Flow for setting up a new 6-digit PIN
quick_login_setup_screen.dart	Settings screen to enable/disable/change PIN and biometric login

4. Splash Screen & App Entry (splash_screen.dart)

The splash screen is the very first screen shown when the app launches. It displays the Vishal Gold brand logo with a fade-in animation, checks the current Firebase Auth session, and automatically routes the user to the correct screen.

Condition	Routed To
No active session	Phone Auth screen (customer OTP login)
Active session — role = USER/WHOLESALER	Customer Home screen
Active session — role = ADMIN/SUPER_ADMIN	Admin Dashboard screen

Admin has PIN enabled	PIN Unlock screen (bypasses full login)
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5. Customer Application — Feature Reference

The customer-facing side is a dark-themed B2B/B2C jewelry catalog and ordering platform. The bottom navigation bar provides access to five main tabs: Home, Search, Cart, Recent Designs, and Profile.

5.1 Home Screen (home_screen.dart)

Component	Description
Banner Carousel	Full-width promotional banners. Each banner is linkable to a URL, product, or category. Managed by admins. Height responsive to screen width.
Category Section	Horizontally scrollable list of all jewelry categories with sub-category tiles below each. Adaptive card width per device.
Recent Designs Strip	Quick-access strip showing the newest product uploads across all categories.
Gold Rate Banner	Displays the current gold rate as set by admin — updates live via Firestore.
Notification Bell	Icon in app bar showing unread notification badge count. Opens Notifications screen.

5.2 Banner Carousel (banner_carousel.dart)

Feature	Detail
Auto-play	4-second interval, wraps continuously
Manual swipe	Left/right swipe gesture supported
Page indicator	Smooth dot indicator (smooth_page_indicator package)
Deep-link actions	Opens external URL, product detail screen, or category/sub-category screen
Responsive height	50% of screen width on phones, scales to 35% on tablets
Data source	Firestore 'banners' collection — real-time stream

5.3 Global Search Screen (global_search_screen.dart)

A full-screen search interface accessible from the bottom navigation bar. Searches across all Categories, Sub-categories, and Products simultaneously.

Feature	Detail
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Search scope	Categories, Sub-categories, Products — all in one result list
Search fields	Category name, sub-category name, product tag number, purity, category display label
Result type badge	Each result shows a colour-coded badge: Category (gold), Sub-type (blue), Product (green)
Leading thumbnail	Product results show product image; categories show icon
Tap action	Category → SubCategories screen; Sub-cat → Product Listing; Product → Product Detail
Auto-focus	Keyboard opens automatically on screen entry
Clear button	X button appears in the search field to clear query instantly
Empty state	Illustrated 'Search for anything' prompt when field is empty
No results state	Friendly 'No results for...' message

5.4 Cart Screen (cart_screen.dart)

The shopping cart. Customers add products from the Product Detail screen. The cart persists within the session and supports quantity adjustments.

Feature	Detail
Cart item list	Each item shows product image, tag number, category, and quantity controls (+/-)
Remove item	Delete button on each item to remove it from cart
Item count summary	Bottom bar shows total item count
Place Order button	Confirms and submits the cart as an order to Firestore
Order confirmation dialog	Shows item count warning before placing; admin is notified
Admin FCM notification	On order placement, a push notification is sent to the admin FCM token
Admin DB notification	A persistent notification record is also stored in Firestore for the admin inbox
WhatsApp alert	After order is placed, a pre-formatted WhatsApp message is sent to the admin support number
Empty state	Illustrated empty cart with 'Browse Designs' button
Post-order routing	Navigates to Order Confirmation screen on success

5.5 Product Listing Screen (product_listing_screen.dart)

Feature	Detail
Grid layout	2 columns on phone, 3 on large phone, 4 on tablet
Skeleton loading	Animated shimmer cards while data loads from Firestore

Staggered animations	Scale + fade-in on first render (flutter_staggered_animations)
Wholesale pricing	WHOLESALE role sees wholesale price; regular users see retail price
Hero animation	Product image animates smoothly between list and detail screens
SliverAppBar	Collapsible header with category/sub-category name and expandedHeight adaptive to screen size

5.6 Product Detail Screen (product_detail_screen.dart)

Feature	Detail
Image gallery	Swipeable multi-image gallery with full-screen zoom support
Full-screen viewer	Tapping an image opens full_screen_photo_viewer.dart with pinch-to-zoom
Product metadata	Tag number, category, sub-category, weight (gross/net), purity, making charges
Role-based pricing	Retail price for customers; wholesale price for WHOLESALE role
Add to Cart	Adds the product to cart with quantity 1; increments if already in cart
Place Sample Order	Navigates to Sample Order screen for custom/reference-based ordering
Share product	Native Android share sheet to send product info
Admin controls	ADMIN/SUPER_ADMIN users see Edit and Delete buttons on this screen

5.7 Full Screen Photo Viewer (full_screen_photo_viewer.dart)

Feature	Detail
Pinch-to-zoom	InteractiveViewer with full zoom support on product images
Swipe to dismiss	Vertical swipe gesture closes the viewer
Dark backdrop	Pure black background for best image contrast
Hero animation	Image hero-animates from product detail into the viewer

5.8 Sample / Custom Order Flow

Screen / Step	Description
category_designs_screen.dart	Browse customer-uploaded reference designs for a given category. Grid layout (2–4 cols). Full-screen image viewer on tap.
sample_order_screen.dart	Multi-step order form: select category & sub-category, enter quantity/size/notes, upload reference images (device gallery OR existing

	Firestore product image), and submit.
order_confirmation_screen.dart	Success screen shown after an order is submitted. Displays the order ID and a summary. Navigates user back to Home.

5.9 Order History Screen (order_history_screen.dart)

Feature	Detail
Order list	All orders placed by the logged-in customer, newest first
Status badge	Colour-coded: Pending (amber), In Progress (blue), Completed (green), Cancelled (red)
Real-time updates	Admin status changes reflect instantly via Firestore stream
Order Detail tap	Tap any order to open full detail with submitted images and notes

5.10 Order Detail Screen (order_detail_screen.dart)

Feature	Detail
Full order info	Order ID, date, category, sub-category, quantity, notes, current status
Image viewer	All reference images submitted with the order shown in a grid; tap to full-screen
Status history	History of status changes with timestamps
Contact admin	Button to contact admin via WhatsApp regarding this specific order

5.11 Recent Designs Screen (recent_designs_screen.dart)

Feature	Detail
Grid layout	2–4 columns based on screen breakpoint
Sort order	Newest products first across all categories
Adaptive padding	Horizontal padding adapts per screen size via AppLayout

5.12 All Sub-Categories Screen (all_subcategories_screen.dart)

Feature	Detail
Grid layout	3–5 columns per breakpoint (3 phone, 4 large phone, 5 tablet)
Sub-cat card	Image thumbnail + sub-category name
Tap action	Opens Product Listing for that sub-category

5.13 Upload Design Screen (upload_design_screen.dart)

Allows users (primarily wholesalers) to upload a new jewelry design to the Firestore catalog for admin review and listing.

Feature	Detail
Form fields	Tag number, gross weight, net weight, name, description
Category selector	Dropdown: 84_ornaments, 92_ornaments, 92_chains
Sub-category	Dynamic dropdown based on selected category
Purity field	84 / 92 KT options
Image picker	Multi-image picker — up to 4 images from device gallery
Upload to Firebase	Images uploaded to Firebase Storage; product doc created in Firestore
WhatsApp notify	Notifies admin via WhatsApp after successful upload

5.14 Company Details Screen (company_details_screen.dart)

Shown to first-time WHOLESALER users immediately after OTP login to collect business details before granting access to the main app.

Field	Description
Company Name	Business / shop name — required
Address	Full business address — required, minimum 5 characters
City	Business city — required
Skip option	Wholesaler may skip and fill in details later from Profile screen

5.15 User Info Screen (user_info_screen.dart)

Displays the logged-in customer's basic profile information in a read-only view. Accessible from in-app links and used as an info card in onboarding contexts.

5.16 Contact Us Screen (contact_us_screen.dart)

Provides customers with the Vishal Gold business contact details including phone number, WhatsApp link, email, and business address. Includes a 'Send WhatsApp' button that pre-fills a message and launches the WhatsApp app.

5.17 Policy Screen / Privacy Policy (policy_screen.dart, privacy_policy_screen.dart)

In-app display of legal documents — Privacy Policy and Terms of Use. Both screens are accessible from the Profile > Settings section. Content is rendered as a scrollable rich-text document.

5.18 Notifications Screen (notifications_screen.dart)

Feature	Detail
Notification list	All notifications for the current user, newest first, from Firestore 'notifications' collection
Unread badge	Gold badge count shown on the notification bell in the Home app bar
Mark all as read	'Mark all as read' button appears in the app bar when there are unread items
Notification types	order_update (order status changes), promotion (flash sales/offers), broadcast (admin messages)
Timestamp	Human-readable relative time shown on each notification tile
Tap action	Tapping a notification navigates to the relevant order or screen
Persistent storage	Notifications stored in Firestore 'notifications' collection; survive app restarts

6. Admin Panel — Feature Reference

The Admin Panel is accessible only to ADMIN and SUPER_ADMIN accounts via email + password login. It uses a dark glassmorphism UI and is divided into four management areas on the dashboard.

6.1 Admin Dashboard (admin_dashboard_screen.dart)

Dashboard Section	Modules / Content
Header (SliverAppBar)	Admin name, avatar, Preview Mode toggle, refresh button, logout button
Quick Actions Strip	Horizontally scrollable fast-access buttons for most-used features
Stats Strip	Live count cards: Products, Categories, Sub-Categories, Admins — pulled from Firestore
Catalog Management	Products, Categories, Subcategories, Banners
Operations	Orders, Sub-Admin Management, CRM Hub, Flash Sale Creator
Intelligence	Analytics Dashboard, Audit Trail, Audit Logs
Tools	Design-to-Social Exporter, FCM Push Console

The module grid is fully responsive: 2 columns on phones, 3 on large phones and foldables, 4 columns on tablets. The SliverAppBar expandedHeight is 200dp on phones, 240dp on tablets.

6.2 Product Management (product_management_screen.dart)

Feature	Description
Create Product	Name, category, sub-category, tag number, weight, purity, making charge %, retail price, wholesale price, multiple images
Edit Product	Modify any field. Changes immediately reflect in customer listings
Delete Product	Removes from Firestore and Firebase Storage
Image Management	Upload multiple images; stored in Firebase Storage with URLs in Firestore
Role-based pricing	Separate retail and wholesale price fields enforced on customer side by role
Gold Rate linking	Prices can be linked to live gold rate for auto-calculation
Preview Mode	Stage changes and preview before publishing to live catalog
Search & Filter	Search by name; filter by category and sub-category
Admin on detail	Admin sees Edit + Delete on the customer-facing Product Detail screen too

6.3 Category Management (category_management_screen.dart)

Feature	Description
Create	Add a new top-level category with name and cover image
Edit	Update category name and image
Delete	Remove category; warns if sub-categories or products exist under it
Reorder	Drag-to-reorder categories to control display sequence in customer app

6.4 Subcategory Management (subcategory_management_screen.dart)

Feature	Description
Create	Add sub-category under a parent category — name + image
Edit	Update name, image, or parent category
Delete	Remove sub-category; optionally re-assign products
Parent linking	Each sub-category linked to parent category via

	Firestore reference
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6.5 Banner Management (banner_management_screen.dart)

Feature	Description
Create	Upload banner image + configure link (URL, product ID, or category)
Activate/Deactivate	Toggle banners on/off. Only active banners appear in the customer carousel
Edit	Update image or link target
Delete	Removes from Firestore and Firebase Storage
Display order	Control sequence of banners in the carousel

6.6 Admin Orders Screen (admin_orders_screen.dart)

Feature	Description
View all orders	All customer orders across all users in a sortable list
Order detail	Customer info, product reference, attached images, notes, timestamps
Status update	Change status: PENDING → IN_PROGRESS → COMPLETED / CANCELLED. Customer sees change in real-time via Firestore stream
Filter by status	Filter orders to focus on PENDING or IN_PROGRESS items
Customer contact	Tap to call or open WhatsApp with customer directly from order detail
Image preview	View customer-uploaded reference images in full-screen

6.7 Sub-Admin Management (sub_admin_management_screen.dart)

SUPER_ADMIN only: This module is hidden from regular ADMIN users.

Feature	Description
Create Admin	Enter name + email; system creates Firebase Auth account and Firestore admin doc
View Admins	List of all active admin accounts with roles and creation dates
Delete Admin	Revokes Firebase Auth + removes Firestore admin document
Role assignment	Assign ADMIN or SUPER_ADMIN role
Permissions	Optionally restrict an admin to specific modules

6.8 CRM Hub (crm_hub_screen.dart)

Feature	Description
Customer Segments	Pre-built live segments with Firestore counts: All Users, Wholesalers, Retail Customers, Recent Customers (last 30 days), Inactive Customers (>90 days no activity)
Push Notification to Segment	Send FCM push notification to any segment. Enter title + body + optional image URL. Delivered to all FCM tokens in the selected segment via Firebase Cloud Messaging
Share Promotions	Upload a custom promotion image from device gallery OR select an existing Firestore product image. Export via native share sheet or Design-to-Social tool
Live Count Dashboard	Count cards for each segment update in real-time as customer registrations change

6.9 Flash Sale Creator (flash_sale_creator_screen.dart)

Feature	Description
Create Flash Sale	Sale name, discount type (% or fixed), eligible product IDs, start/end datetime
Activate / Pause	Enable or pause without deleting
Customer visibility	Active flash sales display discount badge on product cards
Notification trigger	Optionally send push notification to all customers when sale goes live

6.10 Analytics Dashboard (analytics_dashboard_screen.dart)

Metric / Chart	Description
Order Volume	Bar/line chart — orders placed over time (daily, weekly, monthly)
Revenue Trend	Estimated revenue from completed orders over time
Top Categories	Pie chart of most-ordered categories
Customer Growth	Line chart of new user registrations
Wholesaler vs Retail	Order volume split by customer type
Conversion Funnel	App opens → product views → orders placed

6.11 Audit Trail & Audit Logs (audit_trail_screen.dart, audit_logs_screen.dart)

Feature	Description
Audit Trail	High-level recent admin actions grouped by date (product edits, banner changes, admin creation)

Audit Logs	Detailed filterable log — filter by admin, action type, or date range
Log fields	Timestamp, admin name, admin UID, action type, affected entity ID, change summary
Immutable	Log entries cannot be edited or deleted — stored in read-only Firestore collection
Service	audit_service.dart automatically logs every significant admin write action

6.12 Design-to-Social Exporter (design_to_social_screen.dart)

Feature	Description
Template selection	Branded Vishal Gold jewelry promotion templates
Product image embed	Select from Firestore product catalog OR upload from device gallery
Custom text overlay	Add product name, price, or promotional message with styled typography
Export to gallery	Save the finished promotional image to device photo gallery
Share to social media	Native Android share sheet — WhatsApp, Instagram, Facebook, etc.
Brand consistency	All templates use Vishal Gold brand colours and typography

6.13 FCM Push Console (fcm_console_screen.dart)

Feature	Description
Compose	Enter notification title, body text, and optional image URL
Audience selection	All Users, Wholesalers Only, Retail Customers Only, Active Customers
Send immediately	Dispatches FCM notification to all tokens in selected audience
Delivery log	Recent notification log — what was sent, to whom, when
CRM integration	Notifications also triggerable from CRM Hub segment cards

6.14 Admin Preview Mode (preview_provider.dart)

Feature	Description
Stage changes	Draft product/banner edits without publishing to live Firestore
Live preview	Switch app to 'Preview Mode' to see staged changes as a customer
Publish	Push staged changes live with a single confirm action

Discard	Discard all staged changes without affecting live data
Preview banner	A visible banner in the UI tells the admin they are in preview mode

7. Settings & Profile — Shared Features

7.1 Profile Screen (profile_screen.dart)

Feature	Description
Profile avatar	Shows photo from Firebase Storage, or gold initial-letter avatar
Role badge	Gold badge showing USER, WHOLESALER, ADMIN, or SUPER_ADMIN
Account info card	Email, phone, company name (wholesaler), city (wholesaler)
Edit profile link	Pencil icon opens Edit Profile screen
Order History	Quick access link to Order History screen
Logout button	Confirmation dialog then Firebase sign-out + local session clear

7.2 Edit Profile Screen (edit_profile_screen.dart)

Feature	Description
Name update	Change display name; saved locally + to Firestore user document
Profile photo	Pick from device gallery; uploaded to Firebase Storage; URL saved in Firestore
Email update	Optional email field for customers
Company details	Wholesaler can update company name, address, city from this screen

7.3 Settings — Admin Only

Setting Module	Description
Quick Login Settings (quick_login_settings_screen.dart)	Enable/disable PIN; change or reset 6-digit PIN; toggle biometric unlock
Security Center (security_center_screen.dart)	View active sessions, recent login activity, trigger forced sign-out of all sessions
Notification Preferences (notification_settings_screen.dart)	Toggle which system events trigger in-app notifications for the admin
Storage & Data (storage_settings_screen.dart)	View and clear locally cached product images; manage offline storage usage

7.4 Settings — All Users

Setting Module	Description
App Language (language_settings_screen.dart)	Switch app display language. Currently supports English. Framework ready for additional locales.
Support Hub (support_hub_screen.dart)	Contact Vishal Gold support via WhatsApp, email, or in-app link. Shows business hours.
Privacy Policy (privacy_policy_screen.dart)	In-app scrollable Privacy Policy document.
Contact Us (contact_us_screen.dart)	Business phone, WhatsApp, email, and address. 'Send WhatsApp' pre-fills a message.

8. Notifications System

8.1 Notification Types & Triggers

Notification Type	Trigger	Audience
Order Status Update	Admin changes order status in Admin Orders screen	Specific customer (order owner)
New Order Alert	Customer places an order via Cart	Admin (FCM + Firestore DB notification)
Flash Sale Live	Admin activates a Flash Sale	All users (optional push)
Promotional Blast	Admin sends from FCM Console or CRM Hub	Selected segment
New Design Upload	User uploads a design via Upload Design screen	Admin (WhatsApp alert)
Custom Broadcast	Admin composes in FCM Console	Any selected segment

8.2 Notification Storage

Field	Type	Description
id	String	Firestore auto-generated document ID
userId	String	Target user UID (or 'admin' for admin inbox)
title	String	Notification headline
message	String	Notification body text
type	String	order_update, promotion, broadcast, new_order
relatedId	String	Order ID or product ID this notification refers to
isRead	bool	Whether the user has read this notification
createdAt	Timestamp	When the notification was created

9. Background Services & Integrations

Service File	Responsibility
firebase_service.dart	Central Firebase access layer. Wraps all Firestore CRUD, Firebase Storage uploads, FCM token management, banner/product/order streams, support contact lookup, and notification dispatch (sendNotificationRequest, createDbNotification).
firebase_auth_service.dart	Firebase Authentication wrapper — OTP phone auth, email/password auth, session checks, token refresh, and sign-out.
fcm_service.dart	Firebase Cloud Messaging setup — handles incoming push notifications in foreground/background, updates FCM token in Firestore on token refresh, routes notification taps to correct screen.
analytics_service.dart	Firebase Analytics event logging — tracks screen views, order events, search queries, product views, and custom business metrics for the Analytics Dashboard.
audit_service.dart	Writes an immutable log entry to Firestore 'audit_logs' collection on every significant admin action (product/category/banner CRUD, admin creation, order status changes).
whatsapp_service.dart	Utility that constructs and launches WhatsApp deep-links (whatsapp://send or wa.me) with pre-filled messages. Used for: new order admin alerts, design upload alerts, and customer-initiated contact from Contact Us screen.
image_picker_service.dart	Wrapper around the image_picker package. Provides pick-single and pick-multiple methods used across: product management image upload, profile photo update, upload design screen, and CRM promotion image selection.
local_storage_service.dart	Device-local key-value storage (shared_preferences). Stores: user display name cache, app language preference, PIN-enabled flag, and offline product image cache metadata.

10. Database & Infrastructure Overview

10.1 Cloud Firestore Collections

Collection	Purpose	Key Fields
users	Customer & wholesaler profiles	uid, phone, name, role, fcmToken, company_name, address, city, createdAt
admins	Admin accounts	uid, email, name, role, permissions, createdAt
categories	Top-level jewelry categories	name, imageUrl, isActive, displayOrder
subcategories	Sub-categories under categories	name, imageUrl, categoryId, isActive, displayOrder
products	Jewelry product catalog	name, tagNumber, categoryId, subcategoryId, imageUrls[], retailPrice, wholesalePrice, grossWeight, netWeight, purity, makingChargePercent
banners	Home screen promotional banners	imageUrl, linkType, linkValue, isActive, displayOrder
orders	Customer orders (cart and sample)	userId, items[], status, images[], notes, createdAt, updatedAt, customerInfo{}
goldRate	Current gold rate (single doc)	rate, updatedAt, updatedByAdminUid
flashSales	Flash sale configs	name, discountType, discountValue, productIds[], isActive, startAt, endAt
notifications	In-app notification inbox	userId, title, message, type, relatedId, isRead, createdAt
audit_logs	Immutable admin action log	adminUid, adminName, action, entityId, summary, timestamp

10.2 Firebase Storage Bucket Structure

Storage Path	Contents
products/{productId}/	Product images uploaded by admins during product creation/edit
categories/{categoryId}/	Category cover images
subcategories/{id}/	Sub-category thumbnail images
banners/{bannerId}/	Promotional banner images
users/{userId}/profile/	Customer and admin profile pictures
orders/{orderId}/	Customer-uploaded reference images attached to sample orders
designs/{userId}/	Designs uploaded via the Upload Design screen
promotions/{promId}/	Design-to-Social exported promotional artwork

10.3 Firebase Services in Use

Firebase Service	Purpose	Plan
Firebase Authentication	OTP phone auth (customers) + email/password (admins)	Spark (Free)

Cloud Firestore	Primary NoSQL database for all app data	Spark (Free)
Firebase Storage	All images — products, banners, orders, profiles	Spark (Free)
Firebase Cloud Messaging	Push notifications to all user segments	Free
Firebase Analytics	User behaviour tracking and business metrics	Free

UPGRADE NOTE: The Spark (free) plan has usage quotas. Upgrade to the Blaze (pay-as-you-go) plan before going to production to avoid quota blocks on Storage and Firestore reads/writes.

11. Responsive Layout System (lib/utils/app_layout.dart)

A centralised AppLayout utility class adapts all grids, paddings, font sizes, card dimensions, banner heights, and navigation elements across all Android device sizes. Call AppLayout.of(context) from any widget to get device-appropriate values.

Breakpoint	Screen Width	Product Grid	Admin Grid	H-Padding	Stat Tile
Small Phone	< 360dp	2 cols	2 cols	12dp	125dp
Phone	360–599dp	2 cols	2 cols	16dp	125dp
Large Phone	600–839dp	3 cols	3 cols	24dp	150dp
Tablet	≥ 840dp	4 cols	4 cols	32dp	170dp

Screen / Widget	What Adapts
Banner Carousel	Height: 50% screen width on phone → 35% on tablet
Category Section	Card width 115–180dp; list height 155–220dp
All Sub-Categories	Grid columns 3–5+; adaptive horizontal padding
Product Listing	Grid cols 2–4; aspect ratio; skeleton grid; animation column count
Recent Designs	Grid cols 2–4; adaptive padding and spacing
Category Designs	Grid cols 2–4; adaptive padding
Admin Module Grid	Cols 2–4; aspect ratio adapts for tile proportions
Admin SliverAppBar	expandedHeight 200dp (phone) → 240dp (tablet)
Admin Stat Tiles	Width 125–170dp
Admin Body Padding	Horizontal padding 16dp → 32dp on tablet

Authorization & Sign-Off

The undersigned parties confirm that they have reviewed the contents of this document and agree to its terms and findings.

Name:

Signature:

Date:

Client Representative | Vishal Gold

Name:

Signature:

Date:

Delivery Engineer | Engineering Delivery Team